

Solar-to-SAF: Final Model

Cost assessment as of 2026-09-16

Arizona | 100,000 tonnes SAF/year | Solar-to-SAF

Base modeled production cost: \$11.76/kg SAF

Equivalent to \$9.41/litre and \$35.60/US gallon under the stated density assumption. This is a break-even engineering estimate, not an observed SAF market quotation.

Measure	Result
Annual net production cost	\$1,175.7 million
Installed capital	\$9.65 billion
Gross cost before coproducts/support	\$12.32/kg
Net coproduct revenue	\$56.1 million/year
Conventional jet benchmark	\$1.43/kg, observed 2026-09-09
Model premium to that spot benchmark	\$10.32/kg
Weather chronology	8,784 hours; FULL HISTORICAL CALENDAR YEAR

Selected optimal capacities

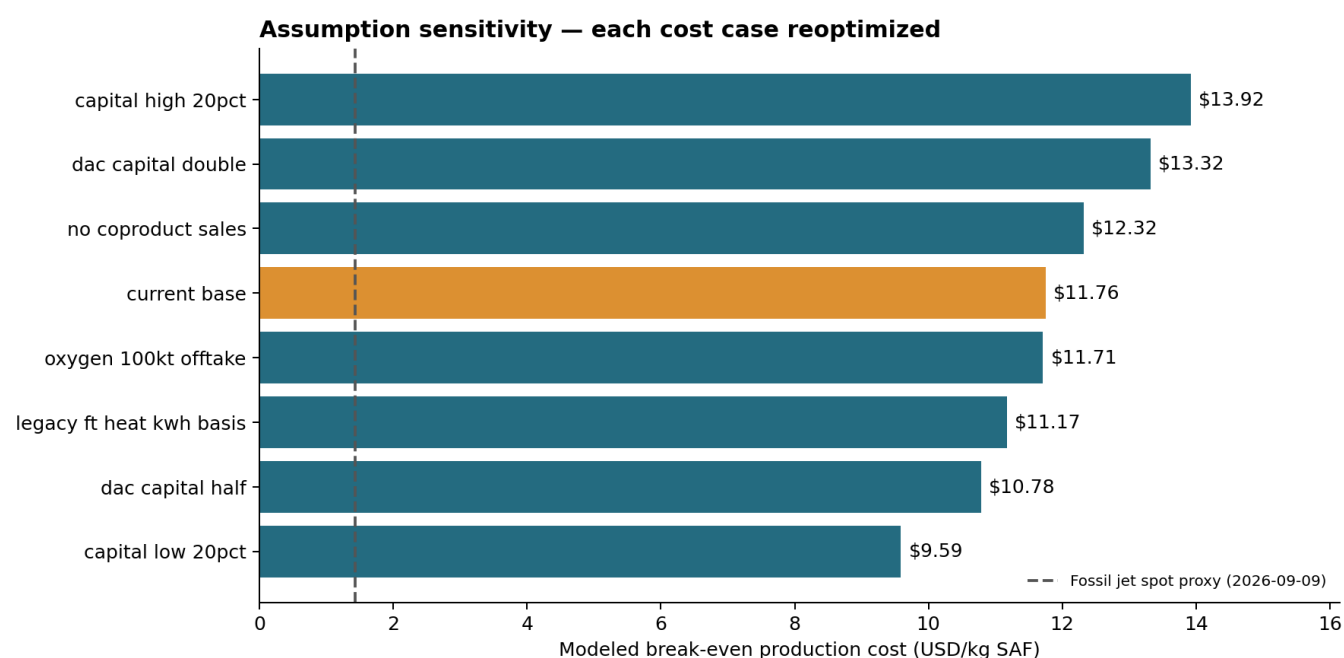
Equipment	Capacity
Solar PV	3.713 GW DC
Electrolyzer	1.162 GW
DAC capture	97.2 tonnes/hour
Battery	611.1 MW / 3929.7 MWh
Hydrogen storage	124.0 tonnes
SAF upgrading	15.12 tonnes/hour

The base case has no oxygen revenue and no policy credit. Liquid coproduct sales use a 90% saleable-volume assumption. Equipment costs combine dated public benchmarks and explicitly identified engineering assumptions.

Sensitivity to assumptions

Scenario	Cost (USD/kg SAF)	Solver status
current base	11.76	optimal
capital low 20pct	9.59	optimal
capital high 20pct	13.92	optimal
dac capital half	10.78	optimal
dac capital double	13.32	optimal
oxygen 100kt offtake	11.71	optimal
no coproduct sales	12.32	optimal
legacy ft heat kwh basis	11.17	optimal

Each equipment-cost or heat case was reoptimized over the full chronology. Offtake changes affect constant revenue only, so those cases reuse the identical optimal design. These cases are conditional scenarios, not a statistical confidence interval. The legacy FT heat case exposes a unit conflict and should not be selected as validated process performance.



Scenarios are conditional estimates, not a statistical confidence interval. Offtake cases retain the same optimal dispatch.

Cost uncertainty remains material. The most valuable next inputs are vendor quotes for DAC, electrolysis and process equipment, a reconciled Aspen heat/mass balance, and actual coproduct offtake terms.

Model checks and assumptions

The base LP contains 87,851 variables and 193,251 constraints. HiGHS returned optimal status in 214.9 seconds. The maximum checked constraint violation is 5.25e-09 in internal MW/MWh/tonne units.

Corrections implemented

- UTC weather timestamps, checked against NSRDB solar geometry; explicit AC output per DC nameplate.
- All auxiliary heat draws electricity and has installed heater capacity; FT heat serves low-temperature DAC only.
- Corrected stated DAC capital units; consistent heat units and explicit uncertainty cases.
- Exact production target, cyclic storage, finite gas-storage flows, battery losses/duration, and cyclic fuel-train ramp limits.
- Bounded coproduct sales, replacement/consumable allowances, zero automatic removal credits, and additive cost exports.

Economic convention and limitations

Historical capital/O&M benchmarks are escalated at an assumed 3%/year to the analysis year, followed by constant real cash flows, a 8% real discount rate and a 25-year life. These are screening choices, not measured inflation or current supplier quotations. EIA observations are short-term fossil-product proxies, not a long-term SAF-price forecast.

The LP optimum is conditional on linear installed costs, fixed surrogate process coefficients, continuous operation and one historical weather year with perfect foresight. It does not certify an Aspen process simulation, lifecycle credit eligibility, fuel specification, plant availability, financing structure or delivered SAF price.

Primary references and reproducibility

[EIA dated petroleum spot-price observations](#)

[DOE 2025 Q1 PV cost benchmarks](#)

[DOE 2024 installed PEM estimate](#)

[IRS carbon-utilization eligibility instructions](#)

See `model_assumptions_used.json` and `market_prices_used.json` for inputs and source statuses; `optimization_summary.json` for recorded versions/hashes and solver checks; `hourly_dispatch.csv` for hourly states; `docs/model_equations.md` for equations; `docs/model_audit.md` for review findings. `docs/file_relocations.json` maps original run paths to the reorganized project files.